

Chaos Programming Language Specification Definition

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Simple Examples

Entry Point and Hello World

```
proc main
  expects argc U64, argv ...String
  returns U64
do
  call print
    with «Hello world!\n»
  return 0
end proc
```

Identifiers and Assignments

```
proc main
  expects argc U64, argv ...String
  returns U64
do
  // These are mutable variables
  let uninitializedVarMut String
  uninitializedVarMut = «String Literal»
  let initializedVarMut String = «String Litral»

  // These are constante variables
  let anotherUninitializedVarConst String
  anohterUnitializedVarConst := «String Literal»
  let initializedVarConst String := «String Literal»

  // Type inference
  let inferedMut = «String Literal»
  let inferedConst := «String Literal»
  return 0
end proc
```

Procedures and Generics

```
// Procedure defintion
proc someCustomProcedure
  expects input1 String, input2 Bool, input3 U64
  returns Void
do
  // perform operations
  call print
    with input1
  return
end proc

// Generic procedure definition
generic T for
```

```

proc someCustomGenericProcedure
  expects input1 T, input2 Bool, input3 U64
  returns Void
do
  // perform operations
  call print
    with input1
  return
end proc

proc main
  expects argc U64, argv ...String
  returns U64
do
  call someCustomProcedure
    with «Hello from Chaos», true, 1
  call someCustomProcedure inserts String
    with «Hello from Chaos», true, 1
  return 0
end proc

```

Structs and Generics

```

proc main
  expects argc U64, argv ...String
  returns U64
do
  // Struct definition
  type CustomStruct struct
    field1 String,
    field2 String = «default value»,
    field3 Bool := true,
  end type
  let customStruct1 CustomStruct :=
    new with
      field1 = «some string»,
      field2 = «another value»,
    end new
  let customStruct2 :=
    new CustomStruct with
      field1 = «some string»,
      field2 = «another value»,
    end new

  // Generic struct definition
  generic T for
  type AnotherCustomType struct
    field1 T,
    field2 String = «default value»,
    field3 Bool := true,
  end type
  let anotherCustomType1 AnotherCustomType :=
    new inserts String with
      field1 = «some string»,
      field2 = «another value»,
    end new
  let anotherCustomType2 :=

```

```
    new AnotherCustomType inserts String with
      field1 = «some string»,
      field2 = «another value»,
    end new
end proc
```

Methods and Generics

Advanced Examples